

⚡ Daily Planner — Make the Counter Buttons

Topic: Wire three buttons to change a number and show it on the page — `getElementById`, `addEventListener`, `count`, `textContent` · **Course:** Web Dev: JavaScript · **Time:** about 50 minutes

You built a Daily Planner page with three buttons: **increase**, **decrease**, and **square**. Now you make them work. Each button changes a **count**, and a text field shows the new number. First you read code and predict, then you fix broken code, then you wire all three buttons yourself.

💬 Words to know: `getElementById`, `addEventListener`, `count`, `textContent`, `square`

1 · Predict 🧠

Read each piece of code. Work out what happens, then write your prediction.

```
let count = 0;
count = count + 1;
count = count + 1;
console.log(count);
```

What number prints?


```
let count = 3;
count = count * count;
console.log(count);
```

What does `count` become? What does "square" mean here?


```
<p id="clicksCounter"></p>

<script>
  const clicksCounter = document.getElementById("clicksCounter");
  clicksCounter.textContent = "Clicks: 5";
</script>
```

The `<p>` started empty. What text shows on the page now?


```
btnInc.addEventListener("click", function () {
  count = count + 1;
  clicksCounter.textContent = "Clicks: " + count;
});
```

The page shows `Clicks: 0`. You click the button three times. What does it show?

2 · Spot the Bug 🐛

Each block is broken. Read what it is **supposed** to do, write the fix, then explain why the original was wrong.

Bug A — This should grab the increase button, whose id in the HTML is `btnInc`. Right now `btn` ends up empty, so nothing works.

```
const btn = document.getElementById("btn");
```

Hint: the id inside `getElementById(" ")` must match the id in the HTML exactly.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Clicking Increase should add **1**. Right now it adds **3** every click.

```
btnInc.addEventListener("click", function () {  
  count = count + 1;  
});  
btnInc.addEventListener("click", function () {  
  count = count + 1;  
});  
btnInc.addEventListener("click", function () {  
  count = count + 1;  
});
```

Hint: how many times did you tell the button to listen? Every listener runs on every click.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The Decrease button should take **1** away from count. Right now clicking it does nothing.

```
const btnDec = document.getElementById("btnDec");  
  
// the increase button is wired, but nothing here talks to btnDec
```

Hint: a button only reacts if you add an `addEventListener` to it.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working **increase** button carefully. It is the pattern you will copy for the other two.

```
<button id="btnInc">increase</button>
<p id="clicksCounter">Count: 0</p>

<script>
  const btnInc = document.getElementById("btnInc");
  const clicksCounter = document.getElementById("clicksCounter");
  let count = 0;

  btnInc.addEventListener("click", function () {
    count = count + 1;
    clicksCounter.textContent = "Count: " + count;
  });
</script>
```

Which line makes the starting count, and what number does it start at?

What does `document.getElementById("btnInc")` give you?

Inside the click function, two things happen every click. Name both, in order.

What does `textContent` change — the code, or the words shown on the page?

4 · Your Assignment: Three Buttons 🛠️

Now make all three buttons work on your Daily Planner. Open your `scripts.js`.

Your page already has these:

```
<button id="btnInc">increase</button>
<button id="btnDec">decrease</button>
<button id="btnSqu">square</button>
<p id="clicksCounter"></p>
```

Make each button change `count`, then show the new number in `clicksCounter`:

- **Increase** — `btnInc` adds 1 to count.
- **Decrease** — `btnDec` takes 1 away from count.
- **Square** — `btnSqu` multiplies count by itself: `count = count * count`.

Start from this. Finish the two `TODO` buttons the same way as increase:

```
const btnInc = document.getElementById("btnInc");
const btnDec = document.getElementById("btnDec");
const btnSqu = document.getElementById("btnSqu");
const clicksCounter = document.getElementById("clicksCounter");
let count = 0;

btnInc.addEventListener("click", function () {
  count = count + 1;
  clicksCounter.textContent = "Count: " + count;
});

// TODO: btnDec — take 1 away from count, then show it
// TODO: btnSqu — multiply count by itself, then show it
```

Write the two listeners you added (decrease and square):

Test it: set count to 3, then click square. What number shows? Click square again — what shows now?

5 · Record Your Walkthrough 🎥

Take a video on a phone while your page runs. Talk like you are teaching someone new. Try to use these words: **button**, **addEventListener**, **count**, **textContent**, **square**.

1. Show your page, then click each button and watch the number change.
2. Read your **square** button's code out loud and say what it does.
3. Point to the line that shows the number on the page.
4. Say what happens to the count when you click square twice.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your video of the three buttons working to teacher on KakaoTalk.